Seminar 5

PROBLEME SERII FOURIER

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Să se dezvolte în serii Fourier următoarele funcţii :

1. $f(x) = \frac{x}{2}, -\pi < x < \pi$

   R: $\frac{x}{2} = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} \sin nx$

2. $$f(t) = \begin{cases} -t - \pi, & \text{dacă } t \in \left[-\pi, -\frac{\pi}{2}\right] \\ t, & \text{dacă } t \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \\ -t + \pi, & \text{dacă } t \in \left[\frac{\pi}{2}, \pi\right] \end{cases}$$

   R: $f(t) = \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{\sin n\frac{\pi}{2}}{n^2} \sin nt$

3. $$f(x) = \begin{cases} 1, & \text{dacă } x \in (0, \pi) \\ -1, & \text{dacă } x \in (\pi, 2\pi) \end{cases}$$

   R: $f(x) = \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{\sin(2n-1)x}{2n-1}$

4. $f(x) = |\cos x|, x \in (-\pi, \pi)$

   R: $f(x) = \frac{2}{\pi} - \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n \cos 2nx}{4n^2 - 1}$

5. $f(x) = |\sin \frac{x}{2}|$

   R: $f(x) = \frac{2}{\pi} \left(1 + 2 \sum_{n=1}^{\infty} \frac{\cos 4nx}{1 - 4n^2}\right)$

6. $f(x) = \sin \alpha x, x \in (-\pi, \pi), \alpha$ nu e întreg

   R: $f(x) = \frac{2 \sin \alpha\pi}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n n}{\alpha^2 - n^2} \sin nx$

7. $$f(t) = \begin{cases} 0, & t \in \left[-\pi, -\frac{\pi}{2}\right] \cup \left[\frac{\pi}{2}, \pi\right] \\ t + \frac{\pi}{2}, & t \in \left[-\frac{\pi}{2}, 0\right] \\ -t + \frac{\pi}{2}, & t \in \left[0, \frac{\pi}{2}\right] \end{cases}$$

   R: $f(t) = \frac{\pi}{8} + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1 - \cos n\frac{\pi}{2}}{n^2} \cos nt$

8. $f(x) = \mathrm{e}^{ax}, a \neq 0$, pe $(-\pi, \pi)$

   R: $\mathrm{e}^{ax} = \frac{2}{\pi} \cdot \mathrm{sh}a\pi \left[\frac{1}{2a} + \sum_{n=1}^{\infty} \frac{(-1)^n}{a^2+n^2} \cdot (a\cos nx - n\sin nx) \right]$

9. $f(x) = |x|$, pe $[-\pi, \pi]$

   R: $|x| = \frac{\pi}{2} - \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{\cos(2n-1)x}{(2n-1)^2}$

10. $f(x) = |\sin x|$, pe $(-\pi, \pi]$

    R: $|\sin x| = \frac{2}{\pi} + \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{\cos 2nx}{1-4n^2}$

11. $$f(x) = \begin{cases} \cos x, & \text{dacă } -\pi < x < -\varphi, \varphi < x < \pi \\ \cos\varphi, & \text{dacă } -\varphi < x < \varphi \end{cases}$$

    R: $f(x) = \frac{1}{\pi}(\varphi\cos\varphi - \sin\varphi) + \frac{1}{\pi}(\pi - \varphi + \sin\varphi \cdot \cos\varphi)\cos x +$
    $+\frac{1}{\pi}\sum_{n=2}^{\infty}\left[\frac{\sin(n+1)\varphi}{n(n+1)} - \frac{\sin(n-1)\varphi}{n(n-1)}\right]\cos nx$

12. Să se dezvolte în serie Fourier de cosinusuri şi apoi de sinusuri funcţia periodică:

$$h(t) = \begin{cases} \frac{x}{4} & \text{pentru } 0 \leq x \leq \frac{\pi}{2} \\ \frac{\pi - x}{4} & \text{pentru } \frac{\pi}{2} < x \leq \pi \end{cases}$$

    R: $f(x) = \frac{\pi}{8} - \frac{2}{\pi} \sum_{k=0}^{\infty} \frac{\cos(4k+2)x}{(4k+2)^2}$

    $f(x) = 2\sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)^2} \sin(2k+1)x$

13. Să se dezvolte în serie de sinusuri pe $(0.\pi)$ funcţia $f(x) = x(\pi - x)$.

    R: $f(x) = \frac{8}{\pi} \sum_{n=1}^{\infty} \frac{\sin(2n-1)x}{(2n-1)^3}$

14. Să se dezvolte în serie de cosinusuri funcţia $f(x) = \frac{1}{2} - \frac{\pi}{4}\sin x$, $x \in [0, \pi]$.

    R: $f(x) = \sum_{n=1}^{\infty} \frac{\cos 2nx}{4n^2-1}, x \in [0, \pi]$

Să se dezvolte în serii Fourier următoarele funcții:

1) $f(x) = \frac{x}{2}$, $-\pi < x < \pi$

Cum f este funcție impară $\Rightarrow$

$$\begin{cases} a_n = 0, \ (\forall) n \geq 0 \\ b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx, \ (\forall) n \geq 1 \end{cases}$$

$$\Rightarrow b_n = \frac{2}{\pi} \int_0^{\pi} \frac{x}{2} \cdot \sin nx \, dx = \frac{1}{\pi} \int_0^{\pi} x \cdot \left(-\frac{\cos nx}{n}\right)' dx$$

integrare prin părți $\frac{1}{\pi} \left(x \cdot \frac{-\cos nx}{n} \Big|_0^{\pi} + \frac{1}{n} \int_0^{\pi} \cos nx \, dx \right) =$

$$= \frac{1}{\pi} \left(\frac{-\pi}{n} \cdot (-1)^n + \frac{1}{n^2} \underbrace{\sin nx \Big|_0^{\pi}}_{=0} \right) = \frac{(-1)^{n+1}}{n} \Rightarrow$$

$$f(x) = \frac{x}{2} = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \cdot \sin nx$$

6) $f(x) = \sin \alpha x$, $x \in (-\pi, \pi)$, α nu este întreg.

Cum f este funcție impară $\Rightarrow$

$$\begin{cases} a_n = 0, \ (\forall) n \geq 0 \\ b_n = \frac{2}{\pi} \int_0^{\pi} \sin \alpha x \sin nx \, dx, \ (\forall) n \geq 1 \end{cases}$$

$$\Rightarrow b_n = \frac{2}{\pi} \int_0^{\pi} \sin(\alpha x) \cdot \sin(nx) \, dx = \frac{2}{\pi} \int_0^{\pi} \frac{1}{2} \left(\cos[(\alpha - n)x] - \cos[(\alpha + n)x] \right) dx =$$

$$\frac{1}{\pi} \int_0^{\pi} \left(\cos[(\alpha - n)x] - \cos[(\alpha + n)x] \right) dx =$$

$$\frac{1}{\pi} \left(\frac{\sin[(\alpha - n)x]}{\alpha - n} - \frac{\sin[(\alpha + n)x]}{\alpha + n} \right) \Big|_0^{\pi} = \frac{1}{\pi} \left(\frac{\sin[(\alpha - n)\pi]}{\alpha - n} - \frac{\sin[(\alpha + n)\pi]}{\alpha + n} \right) =$$

$$\frac{1}{\pi} \left(\frac{\sin \alpha\pi \cdot \cos n\pi - \overbrace{\sin n\pi}^{=0} \cos \alpha\pi}{\alpha - n} - \right.$$

$$\frac{\sin\alpha\pi\cdot\cos n\pi + \cancel{\sin n\pi}^{0}\cos\alpha\pi}{\alpha+n}\Big) = \frac{1}{\pi}\left(\frac{\sin\alpha\pi\cdot(-1)^n}{\alpha-n} - \frac{\sin\alpha\pi\cdot(-1)^n}{\alpha+n}\right)$$

$$= \frac{\sin\alpha\pi\cdot(-1)^n}{\pi}\cdot\frac{\cancel{\alpha}n-\cancel{\alpha}+n}{\alpha^2-n^2} = \frac{\sin\alpha\pi\cdot(-1)^n}{\pi}\cdot\frac{2n}{\alpha^2-n^2}$$

$$\Rightarrow \quad f(x) = \frac{2\sin\alpha\pi}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^n\cdot\sin nx}{\alpha^2-n^2}$$